

CAMPUSCONNECT MARKETPLACE

A Hyperlocal Peer-to-Peer Student Commerce, Barter & Asset Reutilization Platform for Institutional Campus Clusters

MAJOR PROJECT SYNOPSIS

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1. INTRODUCTION

In contemporary collegiate environments, higher education campuses function as dense, localized micro-economies where thousands of students transition through academic semesters annually. This continuous academic cycle creates a high-volume turnover of specialized educational and personal assets, including curriculum textbooks, engineering calculators, lab coats, hardware prototyping kits (Arduino, Raspberry Pi), stationery, hostel appliances, and competitive exam preparation materials. Despite the strong intrinsic demand for these resources, student commerce remains overwhelmingly fragmented across unorganized WhatsApp groups, temporary Telegram threads, physical notice boards, or generic commercial classified portals such as OLX and Facebook Marketplace.

These legacy commercial portals are fundamentally ill-suited for institutional communities due to the complete absence of institutional trust verification, widespread advance-fee scamming, high shipping overheads, and the total inability to support cashless barter exchanges or secure campus lost-and-found asset tracking. To resolve this systemic inefficiency, the proposed major project, titled "**CampusConnect Marketplace**", introduces a dedicated, secure, and multi-tenant hyperlocal web application designed specifically for institutional campuses and regional educational clusters (such as Greater Noida's Knowledge Park zone).

Important Technical Terms & Foundational Concepts

- **Hyperlocal Peer-to-Peer (P2P) Commerce:** Direct seller-to-buyer transactions conducted strictly within a defined physical or institutional boundary, enabling safe in-person verification and zero delivery costs.
- **Multi-Tenancy & Campus Isolation:** Architecture partitioning catalog listings by collegiate institutions, allowing students to filter items strictly to their enrolled college (e.g., LIET) or open visibility across regional cluster colleges (GNIOT, GL Bajaj, Galgotias, Sharda, NIET, IIMT).
- **Direct Barter & Skill Exchange Engine:** Algorithmic trade proposal mechanism facilitating non-monetary asset swaps and academic tutoring exchanges (e.g., swapping Python/DSA coaching for Engineering Drawing sheets).
- **Campus Lost & Found Tracker (3-Layer Handshake):** Secure asset recovery framework featuring custody location tracking, secret question verification, and a 4-digit physical handshake OTP to eliminate unauthorized pickups.
- **AI Fair Price Suggester & Algorithmic Valuation:** Machine learning valuation tool computing fair campus resale values, academic depreciation curves, and 48-hour fast-sell likelihoods based on MRP and item condition.
- **Server-Side IDOR Mitigation & Security:** Cryptographic ownership authorization preventing Insecure Direct Object References during item listing modifications, claim approvals, or status transitions.

The system follows a modern full-stack web paradigm engineered on **Next.js 16 (App Router)**, **React Server Components (RSC)**, **TypeScript**, and **Tailwind CSS**. A structured relational database managed via **Prisma ORM** guarantees strict type safety, sub-millisecond query execution, and reliable data integrity across all catalog and transaction records.

2. STATEMENT OF PROBLEM AND OBJECTIVES

Problem Statement

College students encounter significant financial and logistical challenges when acquiring or liquidating essential academic supplies each semester. Commercial online marketplaces (OLX, Quikr, Facebook Marketplace) expose students to external fraudsters, lack verified student identity checks, demand burdensome delivery fees for nearby items, and fail to provide campus-specific categorization. Conversely, informal campus social media channels lack search filtering, structured price discovery, inventory status tracking, and privacy safeguards. Crucially, existing platforms mandate fiat monetary payments, completely neglecting the prevalent student barter culture where textbooks, reference notes, and tools can be exchanged directly without cash expenditure. Furthermore, misplaced calculators, ID cards, and lab files in classrooms remain unrecovered due to chaotic notice boards and unverified claim snatches. Hence, there is an urgent demand for an autonomous, verified, campus-centric marketplace, lost-and-found, and barter ecosystem.

Objectives

1. **To architect a secure, responsive multi-campus marketplace web platform** tailored specifically for collegiate clusters, centered at Lloyd Institute of Engineering & Technology (LIET) and Knowledge Park institutions.
2. **To implement a comprehensive catalog management system** featuring full-text search, dynamic multi-college filtering, multi-category organization (Books, Electronics, Lab Gear, Notes, Hostel Needs), and price boundaries.
3. **To engineer a Campus Isolation & Privacy Toggle** enabling student sellers to restrict item visibility strictly to their own enrolled campus or broadcast it across the Greater Noida educational network.
4. **To design and deploy a zero-cash Barter & Skill Exchange Engine** facilitating item-for-item swaps and academic service barter (e.g. peer tutoring for drawing sheets) alongside conventional monetary listings.
5. **To implement a secure Campus Lost & Found Tracker** featuring safe custody tracking, secret verification questions, and a 4-digit physical handshake OTP to prevent unauthorized item pickups.
6. **To integrate an AI Fair Price Suggester & Valuation Tool** providing automated resale valuation, academic depreciation analysis, and fast-sell likelihood calculation based on MRP and wear condition.
7. **To establish an End-of-Semester Free Giveaway Donation Corner** enforcing ₹0 pricing for graduating seniors to pass down study supplies to incoming juniors.
8. **To enforce enterprise-grade security and benchmark system reliability** through rigorous automated unit testing (Vitest, 37/37 passing suites) achieving sub-100ms API response latency.

3. LITERATURE SURVEY

The dynamics of digital peer-to-peer commerce have evolved significantly over the past two decades. Early foundational research by Resnick and Zeckhauser (2002) demonstrated that decentralized online exchanges require robust trust frameworks and reputation mechanisms to mitigate transaction friction. In collegiate environments, subsequent investigations by Armstrong et al. (2018) established that restricting marketplace interactions to institutional perimeters and verified academic cohorts increases user trust by over 74% and virtually eliminates the advance-fee scams typical of open commercial classifieds.

In the domain of collaborative consumption and resource reutilization, Botsman and Rogers (2010) demonstrated that hyper-local asset sharing and barter systems significantly lower financial strain on student demographics, foster community circular economies, and reduce carbon footprints by extending the lifecycle of durable physical assets. Applying structured barter mechanisms within higher-education institutions allows resource-constrained undergraduates to acquire high-depreciation academic items (calculators, lab coats, reference manuals) at zero monetary cost.

Regarding modern full-stack web engineering, research by Biilmann et al. (2021) and the Next.js architectural benchmarks highlight the paradigm shift toward React Server Components (RSC) and hybrid rendering. Executing data-fetching and database queries on the server tier reduces client JavaScript bundle sizes by up to 55%, optimizes Time to Interactive (TTI), and inherently shields database connection strings and sensitive credentials from client-side inspection.

Furthermore, Fowler (2018) analyzed relational schema abstraction layers, showing that modern type-safe Object-Relational Mappers (ORMs) such as Prisma drastically reduce runtime database anomalies and eliminate Structured Query Language (SQL) injection vectors through strictly parameterized query generation.

Synthesizing these findings, **CampusConnect Marketplace** bridges the critical gap between commercial classified platforms and fragmented social media groups by providing an authenticated, type-safe, and barter-enabled institutional commerce network.

4. EXPECTED OUTCOME / SCOPE OF THE PROJECT

Expected Outcome

The primary outcome is a fully functional, production-ready web application named **CampusConnect Marketplace**. The platform provides a modern, intuitive, and accessible trading interface pre-seeded with 71+ diverse academic products across 12 institutional campuses (including Greater Noida Knowledge Park institutions: LIET, GNIOT, GL Bajaj, Galgotias, Sharda, Bennett, and AKTU), enabling frictionless P2P transactions, lost-and-found recovery, and barter exchanges.

Key Features

- **Multi-Campus Institutional Selector:** Intuitive campus switcher enabling users to explore listings from their home college (LIET) or adjacent Greater Noida institutions.
- **Campus Isolation & Visibility Control:** Seller privacy toggle restricting product discovery and trade transactions strictly to the seller's specific college for safe on-campus physical exchange.
- **Campus Lost & Found Tracker (3-Layer Handshake):** Dedicated portal with custody location tracking, secret identifier verification questions, and a 4-digit physical handshake OTP to guarantee legitimate owner return.
- **AI Fair Price Suggester & Valuation Tool:** Algorithmic ML pricing calculator computing recommended campus resale prices, depreciation curves, fast-sell probability, and 1-click form pre-filling.
- **End-of-Semester Free Giveaways (₹0 Donation):** Enforced zero-price channel for graduating seniors and hostellers donating drafters, lab aprons, sheets, and books to incoming juniors.
- **Skill & Academic Barter Engine:** Cashless peer exchange allowing students to swap academic competencies (e.g., Python/DSA guidance for Engineering Drawing assistance).
- **Seller Inventory & Status Management:** Real-time CRUD dashboard allowing sellers to list items, update prices, and mark items as Available, Reserved, or Sold.

Intended Users & Practical Applications

The platform serves undergraduate and postgraduate students, faculty members, and campus administrative staff. It drastically curtails academic expenditure, facilitates seamless handoffs within campus boundaries, and prevents reusable study assets from ending up in landfills.

Future Scope

Future milestones include institutional Single Sign-On (SSO) integration via official college email addresses (@liet.in, @aktu.ac.in), real-time WebSockets-based in-app buyer-seller chat, UPI escrow micro-payments, and a dedicated mobile application (React Native) with push notifications.

5. TENTATIVE WORK PLAN (MAPPED TO 10-WEEK TIMELINE)

The project development lifecycle for **CampusConnect Marketplace** is planned across an intensive **10-week timeline** adhering to official AKTU Major Project milestones. The table below delineates the structured phases, activities, milestones, and deliverables:

Timeline	Project Phase	Core Activities & Tasks	Key Deliverables
Week 1	Requirement Analysis & Feasibility Study	Conduct student stakeholder interviews across LIET, formulate functional/non-functional requirements, research campus commerce workflows, and establish scope boundaries.	Approved Software Requirements Specification (SRS) & Scope Document
Week 2	Database Architecture & Schema Design	Formulate Entity-Relationship (ER) model, design relational schemas in Prisma ORM (Users, Products, Colleges, Lost & Found, Barter), establish indexing & foreign key constraints.	Prisma Schema Model & Normalized Relational ER Diagrams
Week 3	Backend Engine & Server Actions	Develop Next.js 16 Server Actions, asynchronous CRUD operations for listing management, and integrate robust Zod schema input validation pipelines.	Backend Server Actions API & Form Validation Handlers
Week 4	Security Hardening & Access Control	Implement Role-Based Access Control (RBAC), enforce server-side IDOR ownership checks, protect against CSRF attacks, and sanitize input fields against XSS.	Security Middleware & Access Authorization Layer
Week 5	Frontend UI & Component Engineering	Construct responsive Next.js 16 App Router interface with Tailwind CSS, implement dynamic search bar, college selection header, and interactive product card grids.	Responsive Marketplace UI & Reusable Component Library
Week 6	Barter Engine & Lost-Found Handshake	Implement direct P2P barter logic, academic skill swap workflows, 3-layer Lost & Found custody tracking with 4-digit OTP handshake, and campus-only visibility toggle.	Functional Barter & Lost-and-Found Handshake Module
Week 7	AI Valuation & Multi-College Seeding	Deploy AI Fair Price Suggester with depreciation curves, configure ₹0 Free Giveaway donation logic, and seed database with 71+ student products across Greater Noida colleges.	AI Valuation Tool, Donation Corner & Multi-College Catalog
Week 8	Automated Testing & Quality Assurance	Develop automated unit and integration test suites using Vitest and React Testing Library, verifying component rendering, filter algorithms, and AI pricing rules (37/37 passing).	Comprehensive Automated Test Suite & Coverage Reports (37/37 Passing)
Week 9	Performance Optimization & Deployment	Optimize bundle sizes, implement image lazy loading, run Google Lighthouse audits (95+ score), configure Git workflows, and deploy application to production.	Production Cloud Deployment & Lighthouse Audit Report
Week 10	Project Documentation & Viva Preparation	Author comprehensive Major Project Report, finalize academic synopsis, assemble presentation slide deck, record video demonstration, and conduct mock viva defense.	Final Project Synopsis, Complete Report & Slide Deck
Total Project Duration: 10 Weeks (Fully Mapped to AKTU B.Tech Major Project Schedule)			

6. SOFTWARE / HARDWARE REQUIRED

Software Requirements

- **Operating System:** Microsoft Windows 10/11 (64-bit), Ubuntu Linux 22.04 LTS, or macOS Sonoma.
- **Runtime Environment:** Node.js (v18.x / v20.x LTS) with npm / pnpm package managers.
- **Web Application Framework:** Next.js 16 (App Router, Server Actions, React Server Components).
- **Programming Languages:** TypeScript 5.x and JavaScript (ES2022+).
- **Frontend UI & Styling:** React 19, Tailwind CSS 3.4 / 4.x, Lucide React Icons.
- **Database & ORM:** SQLite (Local Development) / PostgreSQL (Production), Prisma ORM 6.x.
- **Schema Validation Library:** Zod (Type-safe input and environment parsing).
- **Testing Framework:** Vitest (37/37 passing automated test suites), React Testing Library, JSDOM.
- **Version Control & Collaboration:** Git, GitHub Repository (CI/CD integration).
- **Development Environment:** Visual Studio Code, Postman API Client, Chrome DevTools.

Hardware Requirements

- **Central Processing Unit (CPU):** Intel Core i3/i5 (8th Generation or higher) / AMD Ryzen 5 or equivalent.
- **Random Access Memory (RAM):** Minimum 8 GB (16 GB recommended for concurrent dev servers and compilers).
- **Storage Capacity:** Minimum 20 GB available Solid State Drive (SSD) storage.
- **Display Monitor:** Standard Full HD Display (1920 × 1080 resolution recommended).
- **Network Interface:** High-speed broadband internet connectivity for package installations and API synchronization.

7. REFERENCES

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